Analysis of the edTPA: Instructing and Engaging Students

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Literature Review

What is the edTPA?

The Teacher Performance Assessment (edTPA) is a subject area-specific, performance-based assessment for pre-service teacher candidates, centered on student learning with 25 states are participating. (AACTE, 2013)

Teacher Performance Assessment
Task 1 = Lesson Planning
Task 2 = Implementation of Lesson ← This study
Task 3 = Student Assessment

Task 2

- TCs record 20 minutes of instruction; students collecting, analyzing and interpreting data
- Reflect on this video and write a 3-5 page commentary describing and interpreting the footage
  - Prompted to consider "engage" & "elicit"
- Two rubrics used to evaluated instruction
  - Engaging Students in Learning
  - Deepening Student Learning During Instruction

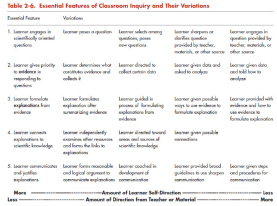
Theoretical Frameworks

Reflective Practice (Schön, 1987)

- Use of video for reflecting on teaching practice

"When someone learns a practice, he is initiated into the traditions of a community of practitioners and the practice world they inhabit. He learns their conventions, constraints, languages, and appreciative systems, their repertoire of exemplars, systematic knowledge, and patterns of knowing-in-action". (p. 36)

Scientific Inquiry (National Research Council, 2000)


Table 2-6. Essential Features of Classroom Inquiry and Their Variations

Essential Feature | Variations

1. Learner engages in scientifically oriented questions. | Learner poses a question | Learner selects among questions, poses new questions | Learner sharpens or clarifies question provided by teacher, materials, or other source | Learner engages in question provided by teacher, materials, or other source

2. Learner gives priority to evidence in responding to questions | Learner determines what constitutes evidence and collects it | Learner directed to collect certain data | Learner given data and asked to analyze | Learner given data and told how to analyze

3. Learner formulates explanations from evidence | Learner formulates explanation after summarizing evidence | Learner guided in process of formulating explanations from evidence | Learner given possible ways to use evidence to formulate explanation | Learner provided with evidence and how to use evidence to formulate explanation

4. Learner connects explanations to scientific knowledge | Learner independently examines other resources and forms the links to explanations | Learner directed toward areas and sources of scientific knowledge | Learner given possible connections

5. Learner communicates and justifies explanations | Learner forms reasonable and logical argument to communicate explanations | Learner coached in development of communication | Learner provided broad guidelines to use to sharpen communication | Learner given steps and procedures for communication

More ——— Amount of Learner Self-Direction ——— Less
Less ——— Amount of Direction from Teacher or Material ——— More

Research Design

Research Questions

• What evidence does the Teacher Performance Assessment reveal about science education teacher candidates' (TC's) understanding of inquiry-based instruction?
•How do the candidates represent their knowledge of inquiry in their reflections?

Qualitative Research Design

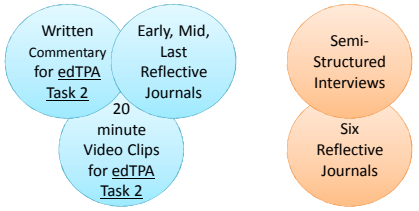
- Case Studies (Yin, 2003)
- Multiple Case-Studies (Stake, 2006)

Context of Study

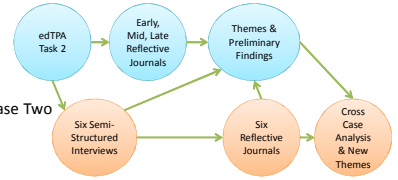
•26 science education teacher candidates
•10 month methods courses
•3rd course in the series
•Concurrent with student teaching
•Common assessment using rubrics

Data Sources and Flow Chart

- Three Data Sources



Phase One



Findings & Implications

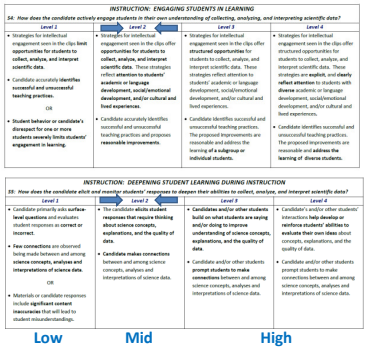
Phase One Data Analysis

edTPA Video Task 2: range of scores of 26 TCs reflected high, mid, low.

- Two Rubrics used to score (SCALE, 2010)

Emergent Themes from edTPA and 3 Reflective Journals:

- Qualities required for Inquiry
  - Teachers and Students
- Definitions of Inquiry
- Barriers to Inquiry
- Misunderstandings of Inquiry


INSTRUCTION: ENGAGING STUDENTS IN LEARNING

Low | Mid | High

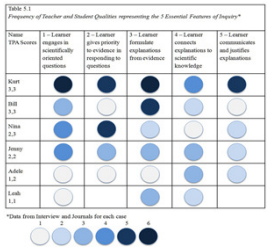
INSTRUCTION: DEEPENING STUDENT LEARNING DURING INSTRUCTION

Low | Mid | High

Phase Two & Cross Case Analysis:

Teacher candidates with a passing score on edTPA Task 2:

- Provided richer descriptions of qualities of inquiry
- Talked about all themes across the ten months.
- Had strong reflection-on-action practice.


Table 5.1
*Frequency of Teacher and Student Qualities representing the 5 Essential Features of Inquiry**

Name TPA Scores	1 – Learner engages in scientifically oriented questions	2 – Learner gives priority to evidence in responding to questions	3 – Learner formulates explanations from evidence	4 – Learner connects explanations to scientific knowledge	5 – Learner communicates and justifies explanations
Kurt 3,3	●	●	●	●	●
Bill 3,3	◐	○	●	◐	○
Nina 2,3	◐	●	○	○	○
Jenny 2,2	◐	◐	○	◐	○
Adele 1,2	○	○	○	◐	○
Leah 1,1	○	○	◐	○	

*Data from Interview and Journals for each case

Quotes from cases that typified their interpretation of inquiry:

"I knew almost nothing about inquiry-based practices before our summer class. I found the topic very exciting, and continue to feel this way." - Kurt

"I'm going to do a lot of classroom discussions." - Bill

"Inquiry… an area where I will need to continue to grow." - Nina

"Inquiry denotes asking questions and seeking answers." - Jenny

"A quiet class is a "boring" class." - Adele

"Cause non-inquiry just has textbooks." - Leah

- All: Originally thought inquiry was hard, now it's easier.
- All but Leah: Inquiry required differentiation.
- K, N, J: Student "frustration" can both be a barrier and good for students.
- Four shared "barriers to inquiry"
  - Time – K, N, J, L
  - Resources – K, B, N, L, A
  - Students – K, N, J, A (B & L had honors students)
  - Novice teacher "hope" – K, N, A



Implications

- Some candidates will need more support.
- Promote the use of language of science education, i.e. "inquiry".
- Promote more in-depth reflections from some candidates.
- Provide cooperating teachers who support inquiry.
- Monitor candidates reflections to observe growth over course, semester, year.
- Need to move beyond teacher-centered concerns to student-centered concerns.

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